Nikhil Kumar

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EXPERIENCE

Edunet Foundation 2024
Web Development Intern Remote
- Designed and developed a responsive personal web application to showcase technical projects and professional achievements using HTML, CSS, and JavaScript.
- Built modular UI components with a focus on scalability, cross-browser compatibility, and mobile-first design.

SRM University AP Jun 2025 – Aug 2025
Research Intern Vijayawada, India
- Designed MATLAB-based simulation framework to study wireless channel effects on MIDI music transmission.
- Modeled AWGN noise, macro jitter, and micro jitter to analyze timing distortions.
- Performed waveform visualization and statistical evaluation to quantify channel impact.

PROJECTS

LemonX – Agentic AI Testing Platform 2026
- Built a full-stack AI-powered testing platform using React, TypeScript, FastAPI, PostgreSQL, and Docker to automate software quality assurance workflows.
- Developed backend APIs for test generation, execution, failure analysis, and automated bug fixing using multi-agent AI systems.
- Integrated GitHub Actions, Vitest, and Cloudflare Workers AI to enable continuous testing, reporting, and automated code repair.

ZKP-Based Medical Trial Eligibility Verification using Blockchain 2025
- Developed a privacy-preserving medical eligibility verification platform using Zero-Knowledge Proofs (Noir), Ethereum smart contracts, and digital signatures.
- Implemented secure credential issuance, proof generation, and on-chain verification workflows while preserving sensitive patient information.
- Built smart contract logic in Solidity and integrated blockchain-based verification mechanisms for decentralized trust.

Brain Tumor Segmentation using Fine-Tuned Segment Anything Model 2025
- Developed a deep learning-based medical image segmentation system using PyTorch and the Segment Anything Model (SAM-ViT-B) for brain tumor MRI analysis.
- Designed data preprocessing, augmentation, training, and evaluation pipelines for large-scale medical imaging datasets.
- Achieved high segmentation accuracy through domain-specific fine-tuning and optimization of foundation models for healthcare applications.

Education

B.Tech in Cybersecurity & Computer Science, SRM AP CGPA: 7.95		2023–2027
Class 12, Sriviswasanthi Education Institution	70%	2022–2023
Class 10, Sriviswasanthi Education Institution	84%	2020–2021

SKILLS

Programming Languages: C, C++, Java, Python, JavaScript, TypeScript, PHP, SQL
Frontend Development: React.js, Next.js, HTML, CSS, Tailwind CSS
Backend Development: FastAPI, Node.js, Express.js, REST APIs, JWT Authentication
Databases: PostgreSQL, MongoDB, MySQL, Firebase
Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, System Design
AI & Machine Learning: RAG, Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow
Tools & Platforms: Git, GitHub, Docker, Linux, GitHub Actions, Postman, Wireshark, Nmap

PUBLICATION

Springer CVR 2026– 6th International Conference on Computer Vision & Robotics
Domain-Aware Adaptation of the Segment Anything Model for Robust Brain Tumor Segmentation in MRI.

ACHIEVEMENTS

- Secured 3rd Runner-Up position in NULLPOINT CTF, demonstrating strong skills in cybersecurity, reverse engineering, and cryptography.
- Selected among the top 3,000 students nationwide for Amazon Summer School 2026, receiving mentorship and advanced training from Amazon scientists in Machine Learning and Deep Learning.

Certifications

Oracle Java Certification	Oracle
PHP Training	Unstop